

SPACECRAFT DYNAMICS CAPSTONE PROJECT

# Attitude Dynamics and Control of a Nano-Satellite Orbiting Mars

COURSE REPORT

STUDENT: Haniyeh Zahra Budaqi

INSTRUCTOR: Prof. Dr. Hanspeter Schaub

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# Abstract

This project presents the development and validation of a complete attitude dynamics and control framework for a nano-satellite operating in low Mars orbit. The mission requires the spacecraft to autonomously transition between three operational pointing modes: Sun-pointing for power generation, nadir-pointing for science observations, and GMO-pointing for communication with a geostationary Mars orbiter.

Reference frames for each mission mode were derived analytically and implemented in MATLAB together with attitude error evaluation and rigid-body rotational dynamics propagation. A proportional-derivative control law was developed to track time-varying reference attitudes and maintain spacecraft stability during mission-mode transitions. The full mission scenario was simulated over a 6500-second orbital timeline with autonomous switching based on illumination and communication visibility conditions.

To complement the numerical simulations, the mission was reconstructed in Systems Tool Kit (STK), providing three-dimensional visual verification of spacecraft orientation throughout the mission. The results demonstrate successful closed-loop stabilization and reliable autonomous transitions between operational modes. Sun-pointing and GMO-pointing objectives were achieved consistently, while the nadir-pointing interval revealed incomplete convergence to the desired Mars-facing orientation. This discrepancy, confirmed by both MATLAB and STK, highlights an area for future controller refinement. Overall, the project validates the integrated attitude-control architecture while demonstrating the value of combining numerical simulation with visual mission reconstruction.

# Chapter 1

## Introduction

This project considers a small science satellite orbiting Mars in a Low Mars Orbit (LMO), tasked with collecting scientific data from the Martian surface and relaying the acquired information to a larger mother spacecraft orbiting in a Geostationary Mars Orbit (GMO). In addition to science and communication operations, the spacecraft must periodically orient its solar panels toward the Sun to maintain sufficient onboard power.

Accordingly, the spacecraft mission is divided into three operational modes:

- 1) **Science Mode:** pointing the sensor toward the Martian surface,
- 2) **Power Mode:** orienting the solar panels toward the Sun,
- 3) **Communication Mode:** pointing the communication antenna toward the GMO spacecraft.

Both the LMO and GMO spacecraft are assumed to follow known circular orbits whose orientations are described using Hill reference frames generated through a 3-1-3 Euler-angle sequence.

The LMO spacecraft is equipped with:

- A science sensor aligned with the  $+b_1$  body axis,
- Solar panels whose normal vector aligns with the  $b_3$  axis,
- A communication antenna aligned with the  $-b_1$  axis.

The objective of this project is to design and validate an attitude determination and control framework capable of autonomously switching between the required operational pointing modes.

## Chapter 2

### Mission Scenario Definitions

#### 2.1 Mission Orbit Overview

The LMO and GMO spacecraft are both assumed to follow circular orbits around Mars. The orbital motion is modeled using classical two-body dynamics, where Mars is treated as the central gravitational body.

The physical and orbital parameters used throughout the simulations are summarized in Table 2.1.

Table 2.1: Simulation and Spacecraft Parameters

| Parameter                                     | Value                    | Unit                    |
|-----------------------------------------------|--------------------------|-------------------------|
| Mars radius $R_{Mars}$                        | 3396.19                  | km                      |
| Mars gravitational parameter $\mu_{Mars}$     | $4.28283 \times 10^{13}$ | $\text{m}^3/\text{s}^2$ |
| LMO altitude                                  | 400                      | km                      |
| LMO orbital radius $r_{LMO}$                  | 3796.19                  | km                      |
| GMO orbital radius $r_{GMO}$                  | 20424.2                  | km                      |
| LMO orbital angular rate $\dot{\theta}_{LMO}$ | 0.000884797              | rad/s                   |
| GMO orbital angular rate $\dot{\theta}_{GMO}$ | 0.0000709003             | rad/s                   |

The orbital angular rates are computed using the circular-orbit relation:

$$\dot{\theta} = \sqrt{\frac{\mu_{Mars}}{r^3}} \quad (2.1)$$

The initial orbit orientation angles of both spacecraft are summarized in Table 2.2.

Table 2.2: Initial Orbit Frame Orientation Angles

| Spacecraft | $\Omega$ (deg) | $i$ (deg) | $\theta(t_0)$ (deg) |
|------------|----------------|-----------|---------------------|
| LMO        | 20             | 30        | 60                  |
| GMO        | 0              | 0         | 250                 |

Figure 2.1 illustrates the definition of a circular orbit using relative elements along with inertial and Hill frames' base vectors. Note that the orbital elements  $(\Omega, i, \theta)$  can be used as (3-1-3) Euler Angles to obtain Hill frame  $\mathcal{H}$  from inertial frame  $\mathcal{N}$ .

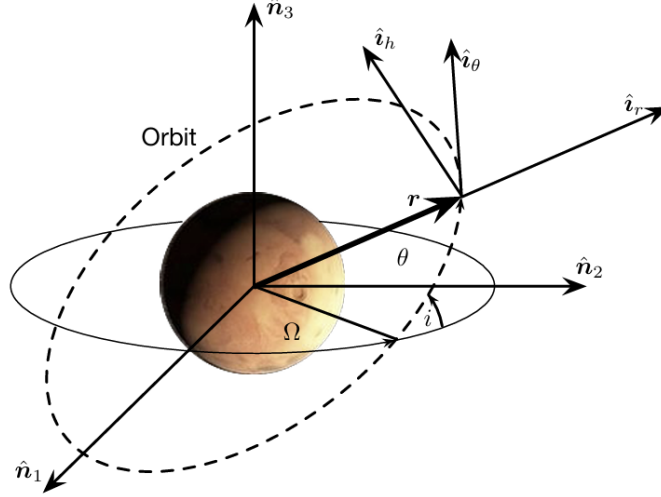


Figure 2.1: Illustration of circular orbital elements alongside inertial frame  $\mathcal{N} : \{\hat{n}_1, \hat{n}_2, \hat{n}_3\}$  and Hill frame  $\mathcal{H} : \{\hat{i}_r, \hat{i}_\theta, \hat{i}_h\}$  base vectors

## 2.2 Mission Pointing Scenarios

A body frame as shown in Figure 2.2 is mounted on the spacecraft, assumed to be the principal body frame.

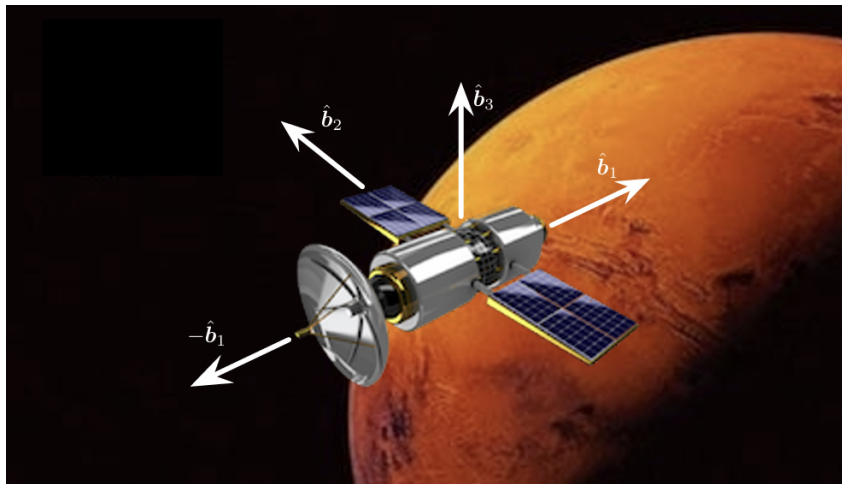


Figure 2.2: Illustration of LMO spacecraft's body frame  $\mathcal{B} : \{\hat{b}_1, \hat{b}_2, \hat{b}_3\}$

Three operational spacecraft pointing modes are defined according to mission requirements.

The Sun is assumed to be located at an effectively infinite distance and aligned with the inertial  $n_2$  axis. Therefore, during Sun-pointing operation, the spacecraft solar-panel axis  $b_3$  must align with the inertial  $n_2$  direction.

The spacecraft operational modes are summarized as follows:

- **Power Mode:** active whenever the spacecraft is on the sunlit side of Mars,
- **Science Mode:** active during eclipse when GMO communication is unavailable,
- **Communication Mode:** active whenever the GMO spacecraft lies within the communication line-of-sight constraint.

The communication mode is activated when the angular separation between the LMO and GMO position vectors is less than  $35^\circ$ .

This operational logic prioritizes spacecraft power generation whenever solar illumination is available, while eclipse periods are allocated to science or communication operations.

Table 2.3: Mission Pointing Scenario Summary

| Orbital Situation                         | Operational Mode   |
|-------------------------------------------|--------------------|
| Spacecraft on sunlit Mars side            | Power Mode         |
| Spacecraft in eclipse and GMO visible     | Communication Mode |
| Spacecraft in eclipse and GMO not visible | Science Mode       |



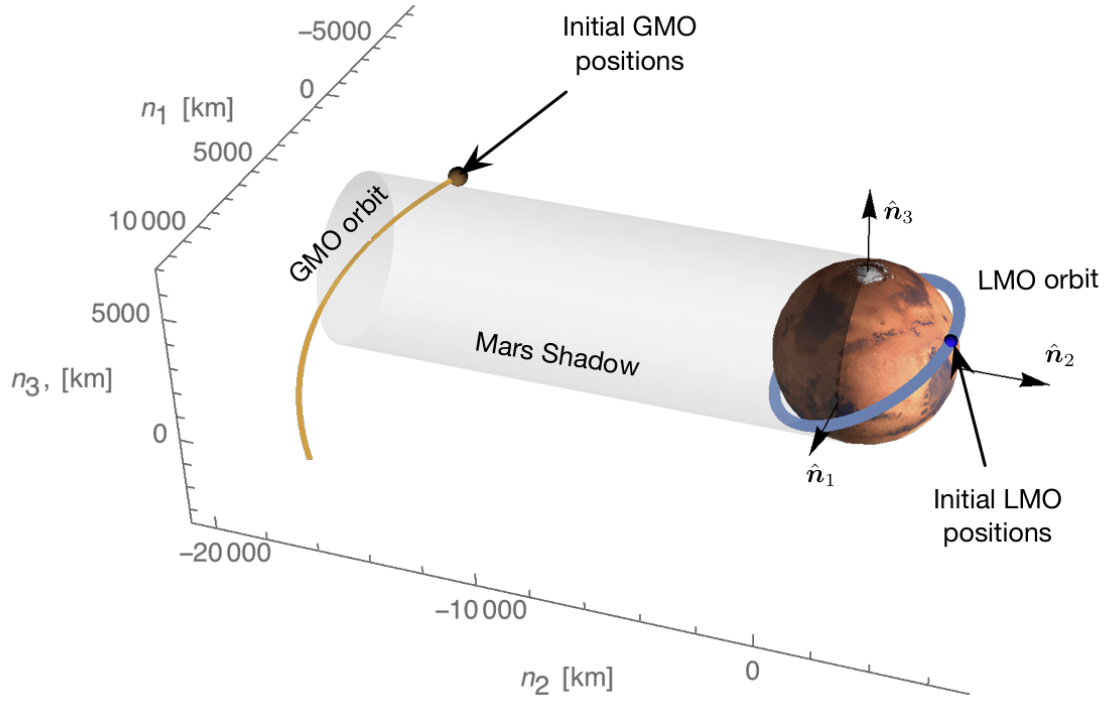


Figure 2.3: Mission Orbit Scenario Illustration

## 2.3 Spacecraft Attitude States

The spacecraft initial attitude is represented using Modified Rodrigues Parameters (MRPs), while the initial angular velocity is expressed in the spacecraft body frame.

The initial spacecraft attitude is defined as:

$$\sigma_{B/N}(t_0) = \begin{bmatrix} 0.3 \\ -0.4 \\ 0.5 \end{bmatrix}$$

The initial body angular velocity is:

$${}^B\omega_{B/N}(t_0) = \begin{bmatrix} 1.00 \\ 1.75 \\ -2.20 \end{bmatrix} \text{ deg/s}$$

The spacecraft inertia tensor is assumed diagonal, implying alignment between the body frame and principal inertia axes.

$${}^B[I] = \begin{bmatrix} 10 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 7.5 \end{bmatrix} \text{ kg}\cdot\text{m}^2$$

The spacecraft is assumed to be capable of generating arbitrary three-axis control torques through an onboard actuator system.

## 2.4 Attitude Control Law Overview

A proportional-derivative (PD) control law is used throughout this project to regulate spacecraft attitude motion:

$${}^B\mathbf{u} = -K\boldsymbol{\sigma}_{B/R} - P{}^B\boldsymbol{\omega}_{B/R} \quad (2.2)$$

where:

- $\boldsymbol{\sigma}_{B/R}$  is the attitude tracking error,
- ${}^B\boldsymbol{\omega}_{B/R}$  is the angular velocity tracking error,
- $K$  and  $P$  are controller gain matrices.

The controller objective is to asymptotically reduce attitude and angular velocity tracking errors while ensuring stable closed-loop rotational behavior.

## 2.5 Implementation Methods

The project was implemented in MATLAB using modular functions for orbital propagation, reference-frame generation, attitude-error evaluation, and closed-loop numerical integration.

The simulation architecture was organized into four primary components:

- Orbit propagation functions to compute LMO and GMO inertial states as functions of time,
- Reference-frame functions to generate Sun-, nadir-, and GMO-pointing DCMs and angular velocities,
- Attitude utilities for MRP/DCM conversion and tracking-error evaluation,
- Mission simulation routines to propagate spacecraft dynamics while automatically switching between operational modes.

A fourth-order Runge–Kutta (RK4) integrator was used for numerical propagation of the attitude state. The modular structure simplified validation of individual tasks before integration into the complete mission scenario.

The final MATLAB results were then reconstructed in STK to visually verify spacecraft orientation and reference-frame tracking throughout the mission timeline.

## Chapter 3

### Task Descriptions and Solutions

#### 3.1 Orbits

##### 3.1.1 Task 1: Orbit Simulation

Assume the general orbit frame as  $\mathcal{O} : \{\hat{i}_r, \hat{i}_\theta, \hat{i}_h\}$ . Even though the general orbit frame and Hill frame basically use the same base vectors, their difference is found in their origin.  $\mathcal{O}$  frame has its origin aligned with the origin of inertial frame  $\mathcal{N}$  – which is the center of Mars. The Hill frame  $\mathcal{H}$ , however, is mounted on the spacecraft, therefore has the same origin as the body frame  $\mathcal{B}$ .

Both these frames are obtainable through a (3-1-3) Euler Angle rotation on the inertial frame, though the hill frame origin is also translated to the spacecraft.

Now, using the  $\mathcal{O}$  frame, spacecraft position vector is expressed as:

$${}^{\mathcal{O}}\mathbf{r} = r\hat{i}_r$$

Both LMO and GMO orbits are circular, therefore having the same angular velocity  $\dot{\theta}$  all over the orbit. Spacecraft velocity vector is expressed as:

$${}^{\mathcal{O}}\dot{\mathbf{r}} = r\dot{\theta}\hat{i}_\theta$$

The MATLAB function `orbitTheta()` takes time, angular velocity and initial angle to calculate the element  $\theta$  using the following equation:

$$\theta_t = \dot{\theta}t + \theta_0$$

Taking the position and velocity vectors from the general orbit frame to the inertial frame is done using the DCM  $[NO]$  which can be obtained having all orbital elements at a given time.

$$[ON] = M_3(\theta_t)M_1(i)M_3(\Omega)$$

$$[NO] = [ON]^T$$

$${}^{\mathcal{N}}\mathbf{r} = [NO] {}^{\mathcal{O}}\mathbf{r}$$

$${}^{\mathcal{N}}\dot{\mathbf{r}} = [NO] {}^{\mathcal{O}}\dot{\mathbf{r}}$$

This is achieved by the MATLAB function `orbit_state()`.

### 3.1.2 Task 2: Orbit Frame Orientation

The Hill-frame basis vectors were constructed directly from the inertial position and velocity vectors:

$${}^{\mathcal{N}}\hat{\mathbf{i}}_r = \frac{\mathbf{r}_{LMO}}{\|\mathbf{r}_{LMO}\|}$$

$${}^{\mathcal{N}}\hat{\mathbf{i}}_h = \frac{\mathbf{r}_{LMO} \times \dot{\mathbf{r}}_{LMO}}{\|\mathbf{r}_{LMO} \times \dot{\mathbf{r}}_{LMO}\|}$$

$${}^{\mathcal{N}}\hat{\mathbf{i}}_\theta = \hat{\mathbf{i}}_h \times \hat{\mathbf{i}}_r$$

To construct a DCM with the base vectors in either frames, we have:

$$[FB] = \begin{bmatrix} {}^{\mathcal{F}}\hat{\mathbf{b}}_1 & {}^{\mathcal{F}}\hat{\mathbf{b}}_2 & {}^{\mathcal{F}}\hat{\mathbf{b}}_3 \end{bmatrix} = \begin{bmatrix} {}^{\mathcal{B}}\hat{\mathbf{f}}_1^T \\ {}^{\mathcal{B}}\hat{\mathbf{f}}_2^T \\ {}^{\mathcal{B}}\hat{\mathbf{f}}_3^T \end{bmatrix}$$

Therefore, to find  $[HN]$  we will have:

$$[HN] = \begin{bmatrix} {}^{\mathcal{N}}\hat{\mathbf{i}}_r^T \\ {}^{\mathcal{N}}\hat{\mathbf{i}}_\theta^T \\ {}^{\mathcal{N}}\hat{\mathbf{i}}_h^T \end{bmatrix}$$

A MATLAB function named `orbit_frame_dcm(t)` was implemented to evaluate the time-varying Hill-frame orientation.

## 3.2 Reference Frame Orientation

### 3.2.1 Task 3: Sun-Pointing Reference Frame Orientation

To point the spacecraft solar panels axis  $\hat{\mathbf{b}}_3$  at the Sun, the reference frame  $\mathcal{R}_s$  must be chosen such that  $\hat{\mathbf{r}}_3$  axis points in the Sun direction ( $\hat{\mathbf{n}}_2$  in the scenario). To define the

frame completely, the first axis  $\hat{\mathbf{r}}_1$  is assumed to point in  $-\hat{\mathbf{n}}_1$  direction. Therefore, the base vectors will be:

$$\begin{aligned}\hat{\mathbf{r}}_1 &= -\hat{\mathbf{n}}_1 \\ \hat{\mathbf{r}}_3 &= \hat{\mathbf{n}}_2 \\ \hat{\mathbf{r}}_2 &= \hat{\mathbf{r}}_3 \times \hat{\mathbf{r}}_1 = -(\hat{\mathbf{n}}_2 \times \hat{\mathbf{n}}_1) = \hat{\mathbf{n}}_3\end{aligned}$$

Similar to the previous task, the analytical expression for the DCM can be obtained using the base vectors:

$$[R_s N] = \begin{bmatrix} \mathcal{N} \hat{\mathbf{r}}_1^T \\ \mathcal{N} \hat{\mathbf{r}}_2^T \\ \mathcal{N} \hat{\mathbf{r}}_3^T \end{bmatrix} = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

The resulting reference-frame DCM can also be obtained using a 3-1-3 Euler-angle of  $(\pi, \frac{\pi}{2}, 0)$  rotation on it. MATLAB function `RsN_DCM()` serves this purpose.

Since the base vectors of this frame are just a fixed permutation of inertial frame basis vectors, the frame remains fixed relative to the inertial frame, thus:

$$\boldsymbol{\omega}_{R_s/N} = 0$$

### 3.2.2 Task 4: Nadir-Pointing Reference Frame Orientation

To point the spacecraft sensor platform axis  $\hat{\mathbf{b}}_1$  towards the center of Mars (nadir direction), the reference frame  $\mathcal{R}_n$  must be chosen such that  $\hat{\mathbf{r}}_1$  axis points towards the planet. To completely describe the frame, we assume the second axis  $\hat{\mathbf{r}}_2$  points in velocity direction  $\hat{\mathbf{i}}_\theta$ . Therefore, we'll have:

$$\begin{aligned}\hat{\mathbf{r}}_1 &= -\hat{\mathbf{i}}_r \\ \hat{\mathbf{r}}_2 &= \hat{\mathbf{i}}_\theta \\ \hat{\mathbf{r}}_3 &= \hat{\mathbf{r}}_1 \times \hat{\mathbf{r}}_2 = -\hat{\mathbf{i}}_h\end{aligned}$$

This frame can be obtained using a (3-1-3) Euler Angles rotation on the Hill frame with angles  $(\pi, \pi, 0)$ .

$$[R_n H] = M_1(\pi)M_3(\pi)$$

We also had previously obtained  $[HN]$  in task 2 as a function of time. So,

$$[R_n N](t) = [R_n H][HN](t)$$

In MATLAB, this task is done by function `RnN_DCM()`.

Since this frame is a re-arrangement of the Hill frame, it also moves with it, thus:

$$\boldsymbol{\omega}_{R_n/N} = \boldsymbol{\omega}_{H/N} = \dot{\theta} \hat{\mathbf{i}}_\theta$$

### 3.2.3 Task 5: GMO-Pointing Reference Frame Orientation

To point the nano-satellite communication platform axis  $\hat{\mathbf{b}}_1$  towards the GMO mother spacecraft, the communication reference frame  $\mathcal{R}_c$  must be chosen such that  $-\hat{\mathbf{r}}_1$  axis points towards the GMO satellite location, defining  $\Delta \mathbf{r} = \mathbf{r}_{GMO} - \mathbf{r}_{LMO}$ . So,

$$\hat{\mathbf{r}}_1 = -\frac{\Delta \mathbf{r}}{\|\Delta \mathbf{r}\|}$$

Further, to fully define a three-dimensional reference frame, assume the second axis is defined as

$$\hat{\mathbf{r}}_2 = \frac{\Delta \mathbf{r} \times \hat{\mathbf{n}}_3}{\|\Delta \mathbf{r} \times \hat{\mathbf{n}}_3\|}$$

The third basis vector is then obtained from

$$\hat{\mathbf{r}}_3 = \hat{\mathbf{r}}_1 \times \hat{\mathbf{r}}_2$$

Having all three base vectors in the inertial frame, we have:

$$[R_c N] = \begin{bmatrix} \mathcal{N} \hat{\mathbf{r}}_1^T \\ \mathcal{N} \hat{\mathbf{r}}_2^T \\ \mathcal{N} \hat{\mathbf{r}}_3^T \end{bmatrix}$$

The MATLAB function `RcN_DCM(t)` is written to deliver this task.

To derive the relative angular velocity  ${}^{\mathcal{N}}\boldsymbol{\omega}_{R_c/N}$ , we'll use an equation (3.27) in the textbook [1] :

$$[\dot{C}] = -[\tilde{\boldsymbol{\omega}}][C] \quad (3.1)$$

Because the reference-frame motion varies smoothly over time, finite-difference differentiation provides a sufficiently accurate approximation of the frame angular velocity. Therefore,

$$[\tilde{\boldsymbol{\omega}}_{R_c/N}] = \begin{bmatrix} 0 & -\omega_3 & \omega_2 \\ \omega_3 & 0 & -\omega_1 \\ -\omega_2 & \omega_1 & 0 \end{bmatrix} = -[R_c \dot{N}][R_c N]^T \quad (3.2)$$

To derive the rate of a DCM numerically, we'll have:

$$[R_c \dot{N}] \approx \frac{[R_c N]_{t+dt} - [R_c N]_t}{dt} \quad (3.3)$$

This algorithm is implemented in MATLAB function `RcN_omega(t)`. The choice of different `dt` will result in different precisions in finding the angular velocity.

### 3.3 Attitude Evaluation and Simulator

#### 3.3.1 Task 6: Attitude Error Evaluation

Any attitude feedback control algorithm requires the ability to compute the attitude and angular velocity tracking errors of the current body frame  $\mathcal{B}$  relative to the reference frame  $\mathcal{R}$ . To find a set of tracking errors  $\begin{bmatrix} \sigma_{B/R} \\ {}^{\mathcal{B}}\omega_{B/R} \end{bmatrix}$ , we have the known variables  $\sigma_{B/N}$ ,  ${}^{\mathcal{B}}\omega_{B/N}$ , as well as the desired reference frame DCM  $[RN]$  and frame rate  ${}^{\mathcal{N}}\omega_{R/N}$ . We first construct the DCM  $[BN]$  which can be obtained from  $\sigma_{B/N}$  using equation (3.152):

$$[C] = [I_{3 \times 3}] + \frac{8[\tilde{\sigma}]^2 - 4(1 - \sigma^2)[\tilde{\sigma}]}{(1 + \sigma^2)^2} \quad (3.4)$$

The inverse transformation—converting a DCM to its MRP form—is formulated in equation (3.153):

$$\begin{aligned} [\tilde{\sigma}] &= \frac{[C]^T - [C]}{\zeta(\zeta + 2)} \\ \zeta &= \sqrt{\text{trace}([C]) + 1} \end{aligned} \quad (3.5)$$

These two forms are implemented in MATLAB function `sigmaToDCM()` and `DCMToSigma()` respectively.

Now, the roadmap to finding the tracking errors would be:

$$\begin{aligned} \sigma_{B/N} &\rightarrow [BN] \\ [BR] &= [BN][NR] = [BN][RN]^T \\ [BR] &\rightarrow \sigma_{B/R} \\ {}^{\mathcal{B}}\omega_{B/R} &= {}^{\mathcal{B}}\omega_{B/N} - {}^{\mathcal{B}}\omega_{R/N} = {}^{\mathcal{B}}\omega_{B/N} - [BN]{}^{\mathcal{N}}\omega_{R/N} \end{aligned}$$

A MATLAB function named `attitudeErrorEval()` was implemented to evaluate both attitude and angular velocity tracking errors.

#### 3.3.2 Task 7: Numerical Attitude Simulator

In order to integrate the attitude we need a numerical integrator. The LMO and GMO orbit satellite positions are already calculated in earlier tasks. Let the propagated attitude

state  $\mathbf{X}$  be

$$\mathbf{X} = \begin{bmatrix} \boldsymbol{\sigma}_{B/N} \\ {}^{\mathcal{B}}\boldsymbol{\omega}_{B/N} \end{bmatrix}$$

Assuming that the spacecraft is rigid and there's no external torque acting on it, its dynamics obey

$$[I]\dot{\boldsymbol{\omega}}_{B/N} = -[\tilde{\boldsymbol{\omega}}_{B/N}][I]\boldsymbol{\omega}_{B/N} + \mathbf{u} \quad (3.6)$$

where  $\mathbf{u}$  is the external control torque vector.

For this task, a loop integrator was written in MATLAB. The control torques  ${}^{\mathcal{B}}\mathbf{u}_1 = 0$  and  ${}^{\mathcal{B}}\mathbf{u}_2 = (0.01, -0.01, 0.02)$  were applied respectively and the following were computed:

- Attitude MRP  $\boldsymbol{\sigma}_{B/N}$
- Angular velocity  $\boldsymbol{\omega}_{B/N}$
- Angular momentum vector  $\mathbf{H} = [I]\boldsymbol{\omega}_{B/N}$  in both  $\mathcal{N}$  and  $\mathcal{B}$  frames
- Rotational kinetic energy  $T = \frac{1}{2}\boldsymbol{\omega}_{B/N}^T [I]\boldsymbol{\omega}_{B/N}$

The results of both simulations are plotted in Figure 3.1. As is seen, after applying the control torque  $\mathbf{u}_2$  on the spacecraft, its rotational kinetic energy increases significantly (from approximately 9 mJ in uncontrolled mode (Figure 3.1a) to more than 2.5 J when controlled (Figure 3.1b)). This is verified by spacecraft angular velocity, whose 1st and 2nd components start to oscillate in controlled mode while its 3rd goes through a rapid, non-stop increase (Figure 3.1d).

The attitude-state histories further confirm this behavior. The spacecraft exhibits unstable rotational behavior when the control torque  $\mathbf{u}_2$  is applied to it (Figure 3.1f). These results indicate that the applied torque does not stabilize the spacecraft attitude dynamics.

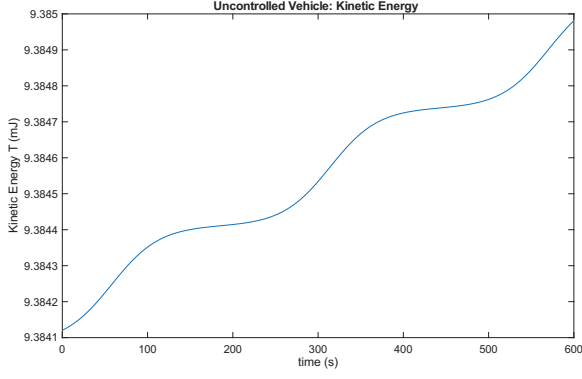
## 3.4 Completing the Mission

### 3.4.1 Task 8: Sun Pointing Control

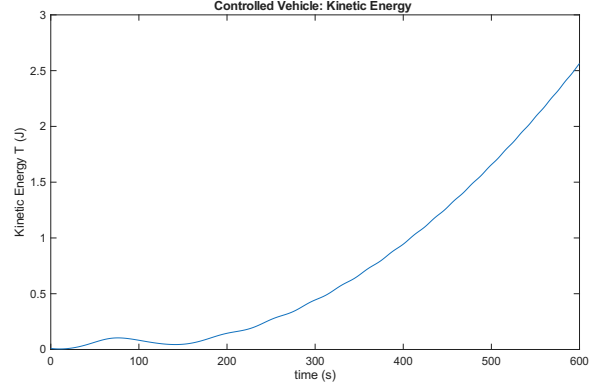
The PD gains were selected using the linearized closed-loop attitude dynamics together with a worst-case inertia design approach. Since scalar gains were used for all spacecraft axes, the gains were chosen based on the maximum and minimum principal moments of inertia to guarantee that all rotational modes satisfy the required 120-second decay time constant.

$$I_{max} = 10, I_{min} = 5$$

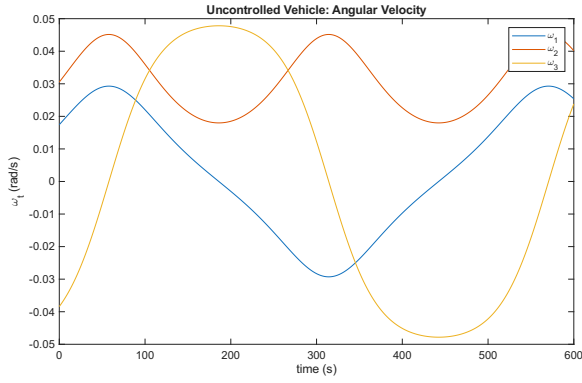




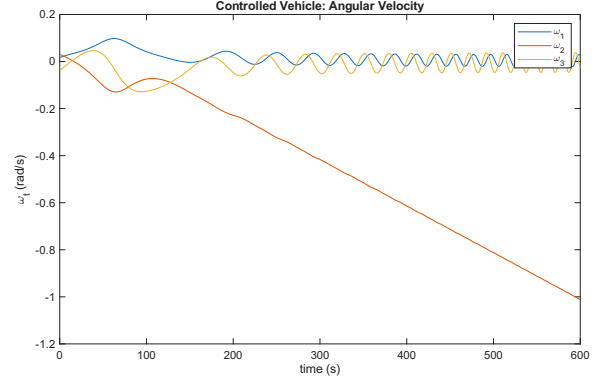
(a)  $u_1$  applied:  $T$



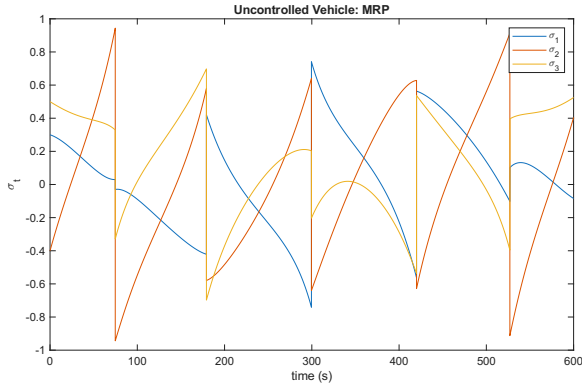
(b)  $u_2$  applied:  $T$



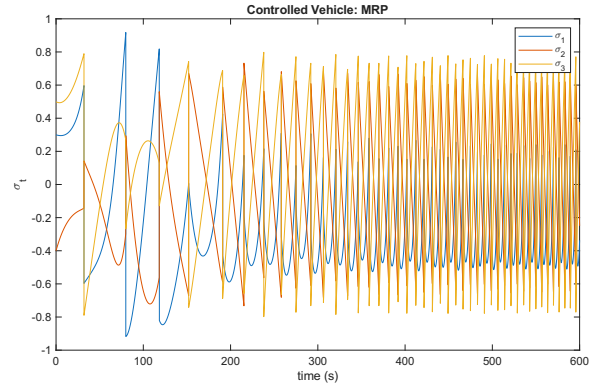
(c)  $u_1$  applied:  $\omega_{B/N}$



(d)  $u_2$  applied:  $\omega_{B/N}$



(e)  $u_1$  applied:  $\sigma_{B/N}$



(f)  $u_2$  applied:  $\sigma_{B/N}$

Figure 3.1: Results of the simulations in task 7

$$\tau = 120s$$

$$K = \frac{4I_{max}^2}{\tau^2 I_{min}}, P = \frac{2I_{max}}{\tau} \quad (3.7)$$

The control torque is defined as:

$${}^B\mathbf{u} = -K\boldsymbol{\sigma}_{B/R} - P{}^B\boldsymbol{\omega}_{B/R}$$

An RK4 integrator was developed to perform numerical propagation of the spacecraft attitude dynamics used by the MATLAB function `simulate_mode()`. The results of Sun pointing simulation are evident in Figure 3.2.

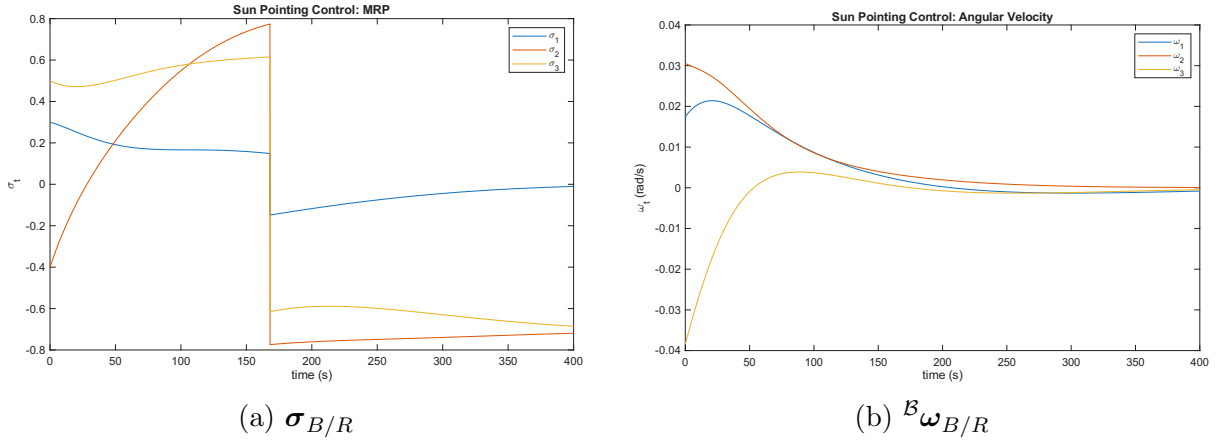


Figure 3.2: Results of Sun pointing control

As is illustrated, the spacecraft is fully detumbled ( ${}^B\boldsymbol{\omega}_{B/R} \approx 0$ ) and driven toward the desired reference attitude.

### 3.4.2 Task 9: Nadir Pointing Control

The same PD control torque developed in task 8 is also applied here and the results of simulation for Nadir pointing mode are evident in Figure 3.3.

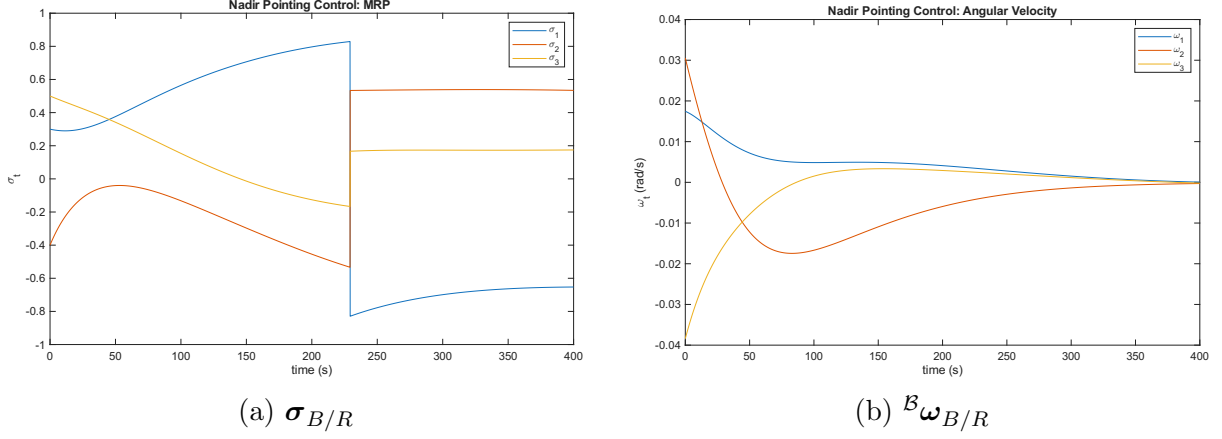


Figure 3.3: Results of Nadir pointing control

Similar to the previous task, the attitude tracking errors converge asymptotically toward zero and the spacecraft is also detumbled.

### 3.4.3 Task 10: GMO Pointing Control

The same PD control torque used in two previous tasks is also applied here and the results of simulation for GMO pointing mode are evident in Figure 3.4.

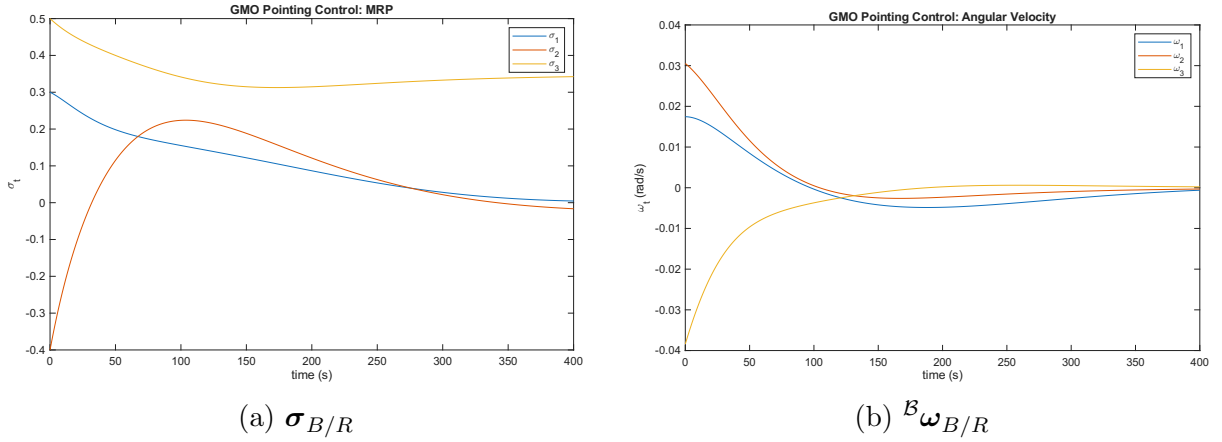


Figure 3.4: Results of GMO pointing control

### 3.4.4 Task 11: Mission Scenario Simulation

A complete mission simulation was developed to automatically switch between spacecraft operational modes according to mission conditions.

The mission logic was implemented as follows:

- If the spacecraft was illuminated by the Sun, Sun-pointing mode was activated,
- If the spacecraft was in eclipse and the GMO spacecraft was visible, communication mode was activated,

- Otherwise, nadir-pointing science mode was selected.

The task of finding out the operation mode at each given time in the mission—and choosing the corresponding reference frame as a result—is carried out by the MATLAB function `getReferenceFrame()`.

The MATLAB function `final_simulation()` handles the procedure of combining all the modes and tasks together in a timespan of 6500 seconds.

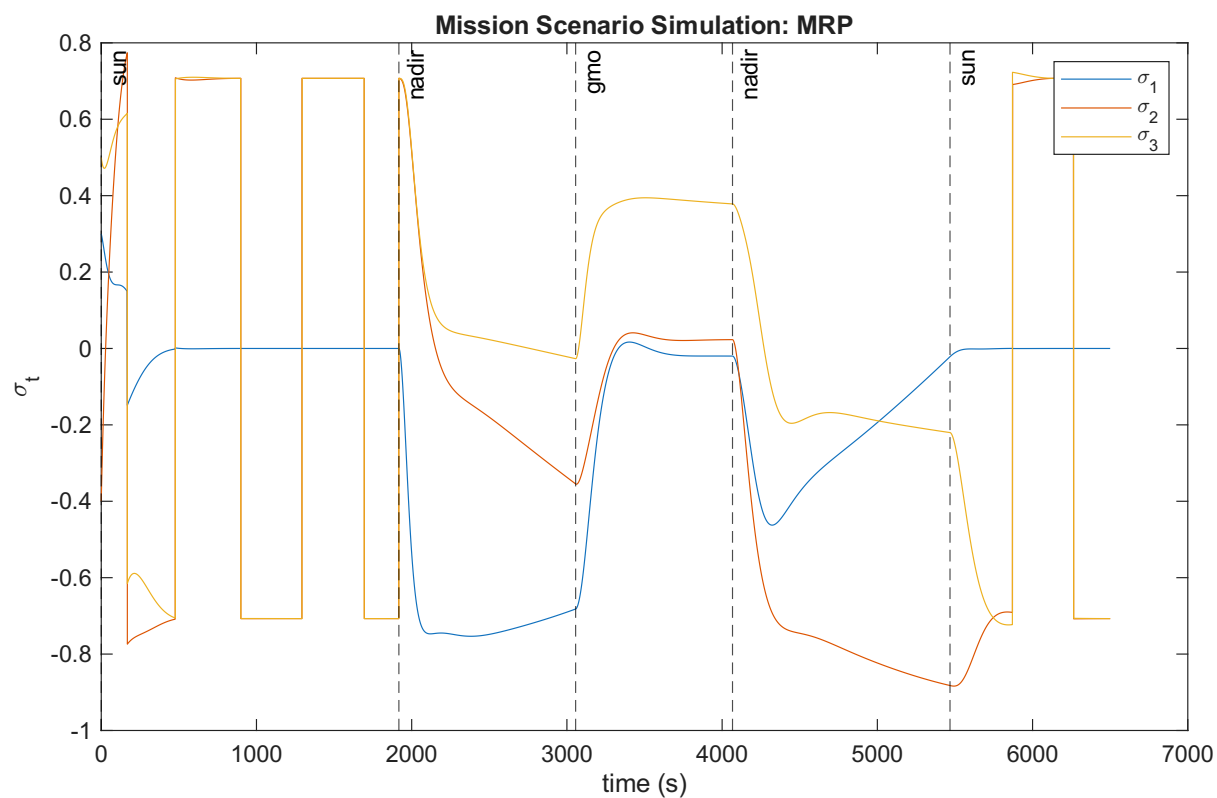


Figure 3.5: Full mission spacecraft attitude state

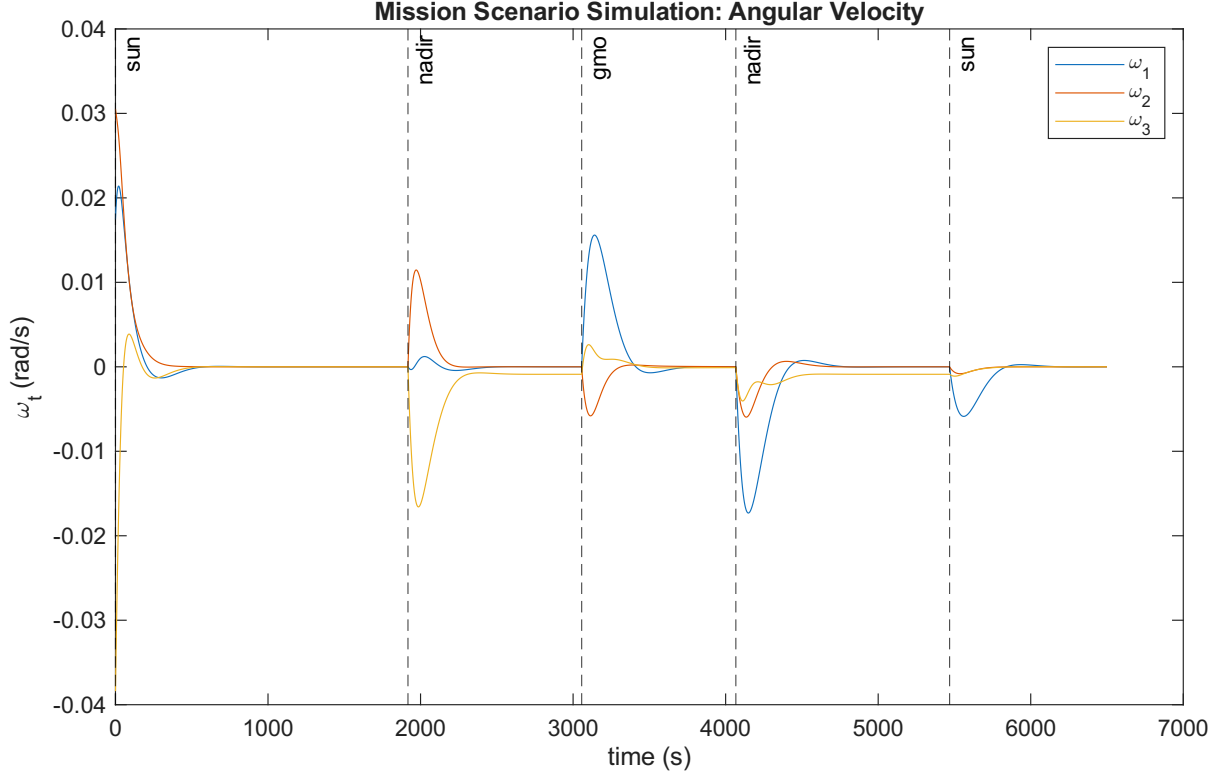


Figure 3.6: Full mission spacecraft angular velocity

### 3.5 Simulation Results and Discussion

The complete mission simulation demonstrates that the spacecraft attitude-control architecture can autonomously transition between multiple operational pointing modes while maintaining stable closed-loop rotational behavior throughout the mission timeline.

Unlike the previous tasks, where each reference frame remained fixed to a single operational objective, the mission scenario in Task 11 requires the commanded reference frame to vary continuously according to orbital geometry and mission conditions. As a result, the controller must track time-varying attitude commands while responding to discrete transitions between Sun-pointing, communication, and nadir-pointing modes.

The numerical results show that the spacecraft remains dynamically stable throughout the simulation and successfully executes each scheduled mission transition. During mode-switching events, temporary increases in attitude and angular velocity tracking errors occur because of the sudden change in commanded reference orientation. These transient responses are short-lived and are effectively damped by the proportional-derivative control law.

The angular velocity histories further demonstrate that the spacecraft remains detumbled throughout the mission despite repeated reference-frame transitions. Brief angular-rate excursions are observed immediately after switching events as the spacecraft reorients toward the newly commanded reference attitude, but no sustained oscillatory behavior

or instability develops.

The simulation also reveals differences in pointing performance between mission modes. Sun-pointing and GMO-pointing exhibit stable convergence and consistent tracking of their commanded orientations. In contrast, the nadir-pointing intervals show slower convergence and incomplete alignment with the commanded Mars-facing reference before the next operational transition occurs. While the controller continues to damp rotational motion and preserve overall stability, the spacecraft does not fully settle into the desired nadir orientation within the available time window.

These results indicate that the implemented control law is effective for autonomous mission execution and spacecraft stabilization, while also highlighting a performance limitation in nadir-pointing operation. This behavior is examined further in the STK visualization chapter, where the three-dimensional mission reconstruction provides additional confirmation of the observed tracking performance.

## Chapter 4

### STK Visualization

To complement the numerical MATLAB simulations, the mission scenario was reconstructed in Systems Tool Kit (STK) to provide visual verification of the spacecraft attitude behavior throughout mission timeline—6500 seconds. The STK visualization enabled direct observation of spacecraft orientation relative to the desired reference frames during each mission segment. This provided an additional validation layer beyond the numerical plots and proved especially useful in identifying attitude tracking performance during transitions between operational modes.

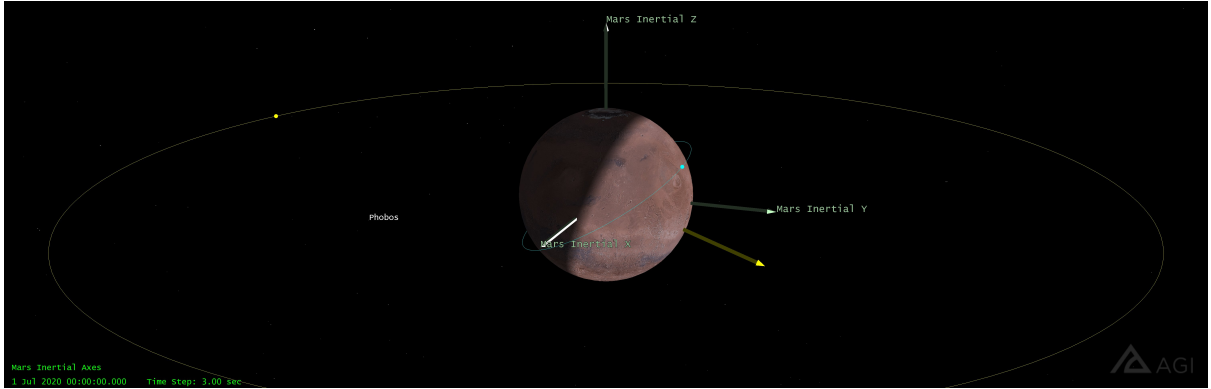


Figure 4.1: Illustration of the central body inertial frame with sun vector (yellow) alongside the spacecrafts' orbits with the blue orbit indicating LMO and the yellow orbit, GMO. The figure depicts the inertial orbital positions of the spacecrafts.

An assumption adopted in the project considered the Sun to be at an effectively infinite distance, with inertial basis vector  $\hat{n}_2$  (in this case, **Mars Inertial Y**) aligned toward it. In real world simulation, however, the date and time 1 July 2020 was the closest to this assumption.

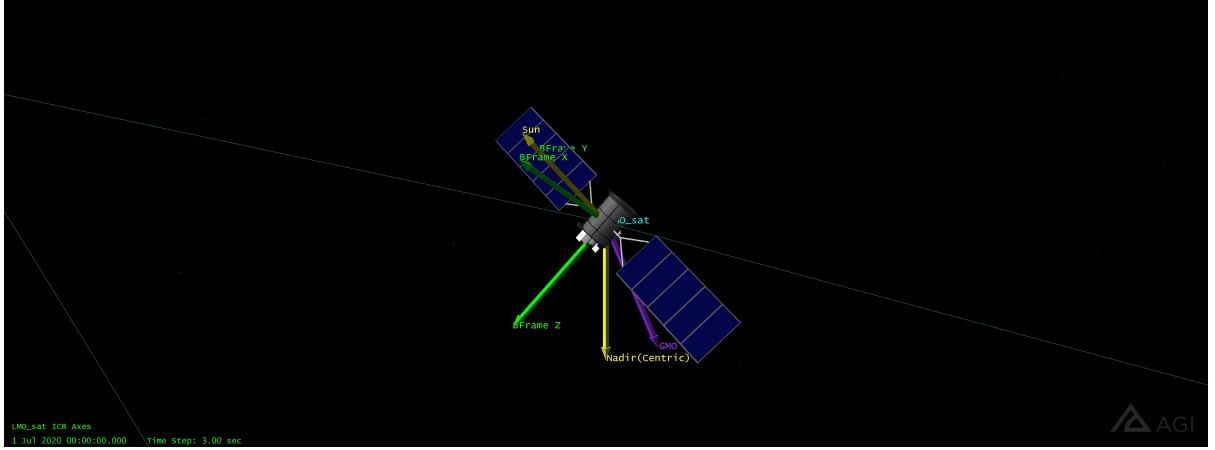


Figure 4.2: LMO spacecraft inertial state, body frame axes and Sun-, Nadir- and GMO-pointing vectors

## 4.1 Sun Pointing Interval

During Sun-pointing intervals, the spacecraft body-fixed reference axis remained aligned with the  $\hat{n}_2$  direction as expected. The STK visualization confirms the behavior observed in the MATLAB attitude history, where the MRP states converge rapidly after the transition and remain bounded throughout the interval. This demonstrates stable tracking performance under the implemented PD control law.

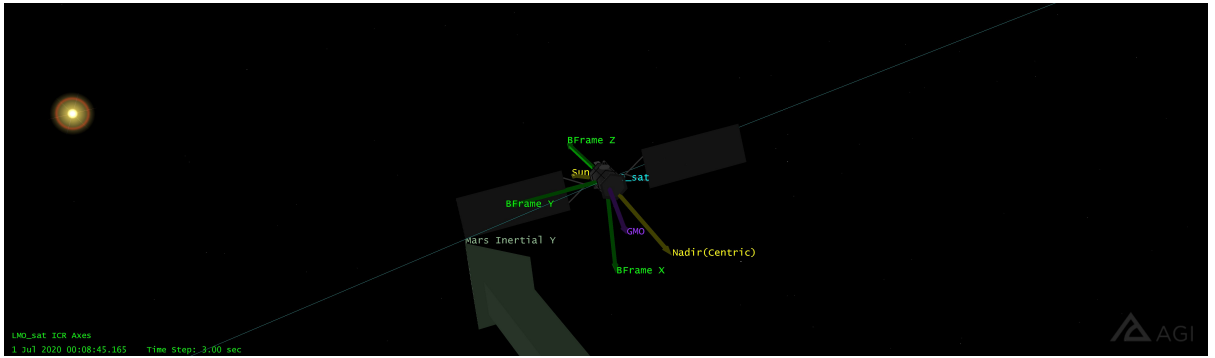


Figure 4.3: STK Sun Pointing Mode

## 4.2 GMO Pointing Interval

During GMO-pointing operation, the spacecraft successfully reorients toward the communication target and maintains a visually consistent line-of-sight throughout the segment. The numerical MRP response shows transient behavior immediately after switching reference frames followed by convergence to bounded values, which agrees with the visual observation in STK.



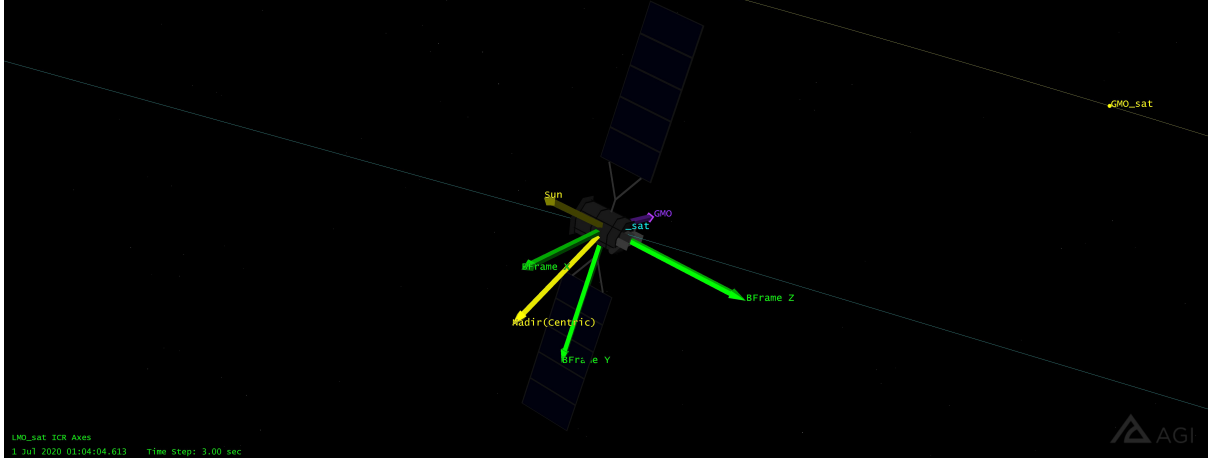


Figure 4.4: STK GMO Pointing Mode

### 4.3 Nadir Pointing Interval

During the nadir-pointing intervals, the spacecraft did not achieve the desired Mars-facing orientation within the available segment duration. This behavior is visible both in the MRP history and in the STK visualization, where the spacecraft attitude remains offset from the expected nadir alignment before the next mission transition occurs. While the controller continues to damp rotational motion and maintains overall stability, the attitude does not fully converge to the commanded reference during this interval.

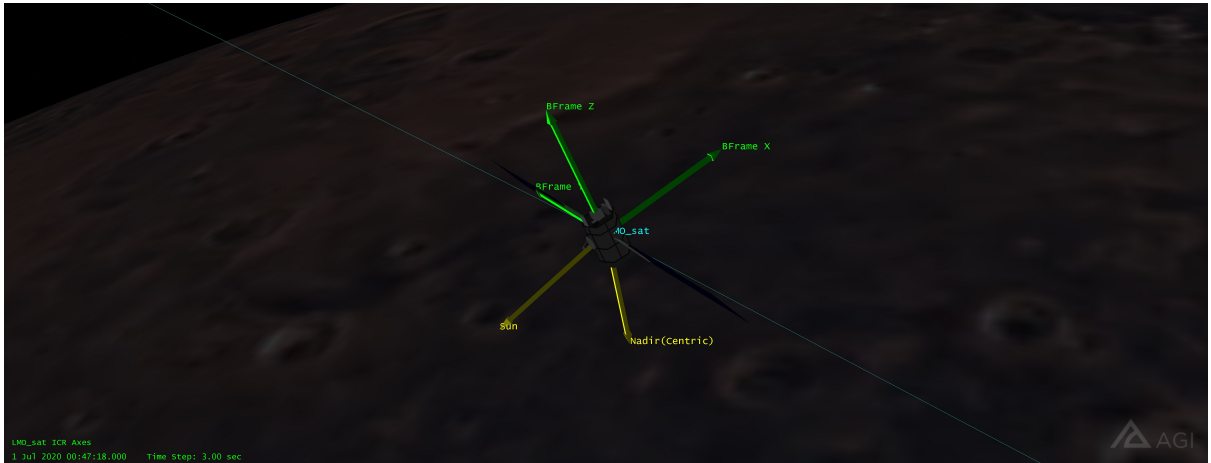


Figure 4.5: STK Nadir Pointing Mode

This discrepancy suggests that the implemented control law provides sufficient de-tumbling performance but may require additional tuning or refinement of the reference tracking formulation for nadir-pointing operation. The STK visualization was especially valuable in identifying this behavior and validating the interpretation of the numerical results.

## 4.4 Mission-level Validation Summary

Overall, the STK reconstruction confirms that the spacecraft remains dynamically stable throughout the mission and successfully transitions between operational modes. Sun-pointing and GMO-pointing behaviors are consistent with the MATLAB simulation results and satisfy the intended mission objectives. Nadir-pointing, however, exhibits incomplete convergence, indicating a limitation in the current controller implementation. The combined MATLAB and STK analysis provides confidence in the numerical model while highlighting areas for future controller refinement.

## Conclusion

This project presented the complete development and implementation of an attitude dynamics and control framework for a nano-satellite operating in low Mars orbit. Mission-specific reference frames were derived for Sun-pointing, nadir-pointing, and GMO communication-pointing configurations and integrated into a unified autonomous mission-management architecture.

The spacecraft rotational dynamics, attitude tracking evaluation, and closed-loop proportional-derivative control system were implemented in MATLAB and tested through numerical simulation. The final mission scenario demonstrated that the spacecraft can autonomously transition between operational modes according to orbital conditions while maintaining stable rotational behavior and remaining detumbled throughout the mission.

To further validate the numerical model, the mission was reconstructed in STK and evaluated through three-dimensional visualization. The STK results closely matched the MATLAB simulations and provided direct visual confirmation of spacecraft orientation during each operational segment. Sun-pointing and GMO-pointing modes successfully achieved their intended objectives and showed strong agreement between numerical and visual results.

A key finding of the project was the performance of the nadir-pointing mode. Although the spacecraft remained dynamically stable and continued damping rotational motion, the commanded Mars-facing orientation was not fully achieved within the available segment duration. This behavior was consistently observed in both MATLAB and STK and suggests that additional controller tuning or refinement of the nadir reference-tracking formulation would improve mission performance.

Overall, the project demonstrates the successful integration of spacecraft attitude dynamics, autonomous control, and mission-level visualization for a multi-mode Mars-orbiting nano-satellite. Future work may include higher-fidelity disturbance modeling, actuator and sensor limitations, smoother slew-transition planning between modes, and more advanced nonlinear control strategies.

## Bibliography

- [1] Hanspeter Schaub and John Junkins. *Analytical Mechanics of Space Systems*. American Institute of Aeronautics and Astronautics, Inc., 2018. ISBN: 9781624105210. DOI: <https://doi.org/10.2514/4.105210>.